



ISO/TC 12, *Quantities and units*

ISO/TC 12 N 1280

2018-10-23

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1. Opening of the meeting (2018-10-08, 09:00)

The chairman welcomed the participants and opened the meeting.

2. Roll call of delegates

12 delegates attended this meeting on site and 4 delegates participated via Zoom, representing **6 P-Members**: Finland, Germany, Japan, Russian Federation, Sweden, and United-States; and **2 Liaison organisations**: BIPM and EURAMET. The **ISO/CS Technical Programme Manager**, Ms. Ferrés was present at the meeting. See N 1273, Attendance list.

Apologies were received from Mr. Curd, UK.

3. Adoption of the agenda

Doc. ISO/TC 12 N 1271

The Agenda was adopted as presented. It was noted that the numbering under agenda item 7 was still not correct. The numbering was redone after the meeting and the correct version reads:

“7.1. ISO/TC 12/WG 19, Revision ISO 80000, all parts

7.1.1 Report of ISO/TC 12/WG 19

7.1.2 FDIS 80000-1, Quantities and units – Part 1: General – Review of issues regarding latest draft and decision on how to proceed

7.1.3 Editors comments on ISO 80000-series

7.1.4 FDIS 80000-1, Quantities and units – Part 1: General – Appointment of Project Leader

7.2 Report of IEC/TC 25/JWG 2, Revision and amendment of IEC-related parts of ISO/IEC 80000 series linked to ISO/TC 12”

4. Appointment of the drafting committee

Ms. Ferrès, Ms. Lindor-Norén and Ms. Saxholm were appointed to assist the secretariat in the drafting of the resolutions.

Resolution 116, Helsinki 1/2018

5. Report of the Chair and of the Secretariat

Doc. ISO/TC 12 N 1261 (Chairman's report) N 1252 (report of the secretariat)

The chair, Prof. Pendrill, presented doc N 1261, Chairman's report to the plenary meeting. He pointed out that there are three bodies active in the field, the JCGM (responsible for VIM, Vocabulary of International Metrology, CCU (responsible for the SI brochure) and ISO/TC 12. Much of the work of TC12 with the ISO 80000-series (particularly part 1) in the past year had benefitted from extensive mutual contacts and exchanges between TC12 and the other two bodies.

He gave an overview of recent trends in science and technology regarding quantities and units. One such sector is the health care sector where ISO/TC 12 has an active collaboration with ISO/TC 215, Health informatics (see also items 8 and 9).



The secretary presented the report of the secretariat, N 1263.

The TPM commented that the member bodies should be invited to participate in a positive manner.

6. Business Plan for ISO/TC 12

Doc. ISO/TC 12 N 1253

The business plan was discussed. The Finnish delegate Mr. Kurko-Suonio raised a question on 2.1.9 6th paragraph. (He has provided a detailed meeting feedback separately.) The chair agreed with the Finnish concerns and encouraged the Finnish delegation (as well as all other delegations) to prepare a text for this item. Dr. Tanaka expressed his support for the Finnish view and emphasised that education is an important part of dissemination which is a key action of the committee.

Resolution 117, Helsinki 2/2018

7. Reports of the working groups

7.1 ISO/TC 12/WG 19, Revision ISO 80000, all parts

7.1.1 Report of WG 19

Doc. ISO/TC 12 N 1254

The secretary of WG 19, Mr. Dominik, reported on the activities of the working group since the last ISO/TC 12 meeting. All documents have been completed and have been forwarded to the secretariat for final editing. Because the work in WG 19 now is completed the Working Group will be disbanded.

ISO/TC 12 thanked Mr Dominik and the members of WG 19 for their dedicated work.

Resolution 120, Helsinki 5/2018

7.1.2 FDIS 80000-1, Quantities and units – Part 1: General – Review of issues regarding latest draft and decision on how to proceed

Doc. ISO/TC 12 N 1220, N 1248, N 1255

The secretary presented doc, N 1255, Review of changes in the draft for FDIS 80000-1.

The chair elaborated on the changes. Referring to 1.1 in the list,

Item 1 the sentence regarding ISQ.

Item 2, he considered it being premature to delete the definitions for ordinal quantity and nominal property.

Item 3, "Dimension of quantity number" needs to be aligned by the other bodies.

Ms. Estafania, BIPM reported on the progress of the 9th edition of the SI brochure. The 9th edition has now been revised. The SI brochure will be published in November 2018. Remaining issues will be readdressed in the work on the 10th edition. The SI brochure does not deal with ordinal quantities and nominal properties.

Dr. Knopf noted that the issue of ordinal quantity and nominal property should not be decided at this meeting. The chair noted that VIM is expected to deal with this issue in the near future.

ISO/TC 12 decided to cancel the ongoing revision of ISO 80000-1:2009.



The members present nominated Prof. Pendrill to take the role as project leader for the continued work on ISO 80000-1 (Germany abstaining). Prof. Pendrill noted that the work needs to be finished in the near future.

ISO/TC 12 decided to initiate a new revision of ISO 80000-1:2009, under the Vienna Agreement, ISO lead, the scope remains unchanged, the development time frame is 36 months CD ballot initiated 2019-03-30 and the project leader to be Prof. Pendrill.

Resolution 118, Helsinki 5/2018
Resolution 119, Helsinki 4/2018

7.1.3 Editors comments on ISO 80000-series **Doc. ISO/TC 12 N 1267**

Ms. Ferrès presented doc. N 1267. All ISO standards are converted to HTML. Ms. Ferrès has noted that in the foreword to the standards, the changes compared to the previous edition are missing. The project leaders will be asked to provide this information.

The scopes need to be harmonised. An issue is whether to limit the ISO 80000-series to only SI units or not.

In the tables, the item numbers from previous edition were discussed. Mr. Kurko-Suonio stressed that it important not to limit the units to SI units. He was also in favour of keeping the reference to the previous editions.

The names of the units are given in the SI brochure. The SI brochure will be referenced in the standards.

The issue of "speed" and "velocity" was discussed. "Speed" and "velocity" are not synonyms in English and the text will be updated accordingly. Mr. Kurko-Suonio pointed out that this is the only case in ISO/TC 12 where the vector quantity and the scalar quantity have different names. The mathematics will be reviewed.

Prof. Jacobson proposed that these issues should be resolved.

ISO/TC 12 thanked the ISO/CS for their assistance and work on the documents.

In response to a need for an editorial committee for consistency issues in parts 2 - 12, Prof. Grimvall and Prof. Pendrill agreed to assist the ISO Central Secretariat editor in the finalisation of the ISO 80000-series for FDIS ballot.

ISO/TC 12 noted that Prof. Grimvall is the project leader for IEC 80000-6, Electromagnetism.

Mr. Kurko-Suonio observed that there is a need to have a bridge between property and quantity.

The writing rules need to be revisited regarding the order of the units.

7.2 Report of IEC/TC 25/JWG 2, Revision and amendment of IEC-related parts of ISO/IEC 80000 series linked to ISO/TC 12

Doc. ISO/TC 12 N 1247 (IEC/CD 80000-6, Electromagnetism) **and N 1256** (Report of voting on IEC/CD 80000-6, *Electromagnetism*)

Prof. Jacobson informed that Prof. Obrzalek has been nominated to be convenor of the JWG 2. ISO/TC 12 will circulate the proposal for a 4 week CIB.



8. Report on the work of ISO/TC 215 TF1, Task Force on Quantities and Units to be used in e-health Doc. ISO/TC 12 N 1234

The chair reported on the Work of ISO/TC 215 TF. Apart from the general ISO/TC 215 report, the latest draft version of the TF's recommendations is expected to be available later this year. The chairman welcomed feedback on this document when it becomes available.

9. Definition of the term measurement

Doc. ISO/TC 12 N 1268, N 1265

The chair reported on the issue of the definition of the term "measurement". The VIM committee will address this issue in the upcoming discussions for a new edition of VIM. ISO/TC 12 mentions nominal and ordinal properties in its current business plan, but is waiting for the progress in JCGM and will give input to the issue once an ISO/TC 12 position has been formed.

ISO/TC 12 did not yet have a position on this.

Ms. Ferrès will participate in the next meeting of JGCM.

10. Review of liaisons

10.1 ISO committees in liaison

JTC 1/SC 32, TC 8/SC 6, TC 37, TC 43, TC 43/SC 1, TC 43/SC 3, TC 47, TC 69, TC 85, TC 115, TC 203, TC 215

Doc. ISO/TC 12 N 1263

ISO/TC 215 has provided a report of activities. The main items have been dealt with under Item 8.

10.2 IEC committees in liaison

TC 1, TC 25

Doc. ISO/TC 12 N 1262

The secretary presented doc. N 1262, liaison report to ISO/TC 12 Quantities and units, from IEC/TC 25, Quantities and units.

10.3 ITU-T/SG 17 – Security

Doc. ISO/TC 12 N 1264

Ms. Ferrès presented doc. N 1264 on behalf of Mr. Caras. The chair and secretary will contact ITU-T/SG 17 for further coordination and collaboration on the new work item proposal on "Biology to Machine Protocol".

Resolution 121, Helsinki 6/2018

10.4 Organizations in liaison

BIPM, CIE, EURAMET, ICRU, ILO, IMO, Infoterm, IUCr, IUPAC, IUPAP, OASIS, OIML, UNECE, WHO, WMO

Doc. ISO/TC 12 N 1266

The chair presented doc. N 1266 on behalf of Mr. Zwick.

Mr. Jarvis reported on EMPIR, the European Metrology Programme for Innovation and Research. So far 17 pre-normative projects under EMPIR are running with 8 projects initiated by needs directly expressed by the CEN or CENELEC standardisation group. A short description of these European projects can be consulted at: <https://www.euramet.org/research-innovation/research->



[empir/empir-calls-and-projects/](#) the possibility to address items for future standardisation needs that need research.

10.5 Review of list of liaison representatives from ISO/TC 12 to liaisons and decision on changes

Doc. ISO/TC 12 N 1252

The secretariat was tasked with contacting the organisations which have not yet appointed liaison representatives to ISO/TC 12.

11. Work items on which no progress is being made - Status and action to be taken

Despite some progress, ISO/TC 12 resolved to abandon the present project for ISO 80000-1 and to open a new project on this issue, see agenda item 7.1.2.

12. Items for future work

12.1 ISO/TC 19 – Preferred numbers

Doc. ISO/TC 12 N 1250

Ms. Ferrès presented document doc. N 1250. ISO/TC 12 resolved that there would be no actions taken at this time.

12.2 Data representation for quantities and units

Doc. ISO/TC 12 N 1258

The chair presented doc. N 1258. He reported the UNECE has set up a codification for quantities and units. This has also emerged as a relatively major issues in the work of the ISO TC215/12 TF Quantities & units:e-health (item 8) .The chair recommended that we should collaborate with organisations who do work on this.

ISO/TC 12 resolved postpone the item until the next meeting, to have a more detailed discussion at the next meeting.

Resolution 122, Helsinki 7/2018

12.3 IEC 80000-18, Quantities and units – Part 18: Software product

Doc. ISO/TC 12 N 1260 and N 1270 (Result of Voting)

The chair presented doc. N 1260. Prof. Jacobson commented that the document was rejected by the German National committee. The lay out and content is to different from the other parts in the ISO and IEC 80000-series. Dr. Tanaka reported that Japan also voted against the proposal.

Germany has reported that they have changed their vote after the voting closed. This means that the document was disapproved in ISO/TC 12.

13. Requirements concerning a subsequent meeting

The next meeting will be held in the last week in September 2019 in The Russian Federation.

Resolution 123, Helsinki 8/2018

14. Any other business

No other business.



15. Approval of resolutions

The resolutions were read out loud and approved.

16. Closure of the meeting (17:00)

The chairman expressed the committees thanks to The Finnish Standards Association for hosting the meeting and the kind assistance of Kati Heiskanen, Sirpa Sipola and Timo Kangas.